

IEEE 14-Bus N-1 Contingency Analysis and Grid-Risk Dashboard

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Objective and model

A reproducible Python/pandapower workflow solves the 14-bus, 20-branch benchmark; validates the AC base case; trips every line and transformer; detects islanding; checks voltage and scenario thermal criteria; and ranks events with a traceable score. The network supplies 259.0 MW through 15 lines and five transformers.

13.393 MW

Base losses

20

Outages

19

AC converged

1

Islands

Validation gate

Newton-Raphson and fast-decoupled AC both converged. Maximum voltage disagreement was 7.12e-10 p.u.; maximum angle disagreement was 3.74e-08 degree. Generation - demand - real branch losses closed to 4.97e-14 MW. The outage engine runs only after these checks pass.

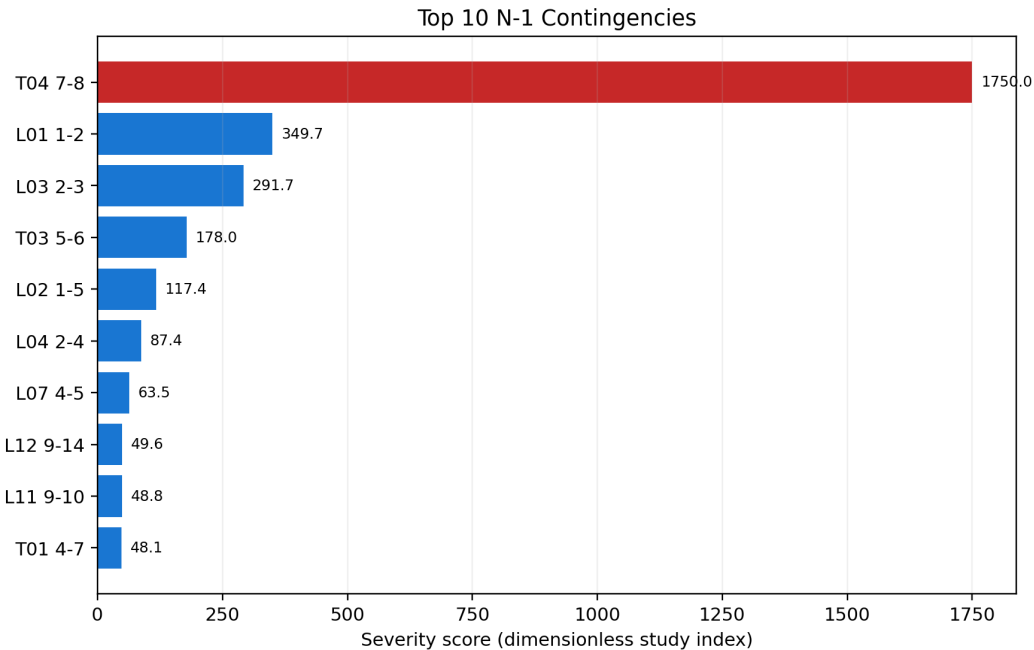
Criteria and transparent score

Voltage uses the case limits (0.94-1.06 p.u.). The source case supplies no branch rateA values, so thermal percentages use a disclosed stress-test rating: ceil-to-5(max(40 MVA, 1.25 x base terminal MVA)). These are not utility equipment limits.

Severity = 1000(islanding) + 750(no valid full-system AC solution) + 100(unsupplied-load fraction) + 1000(total voltage breach p.u.) + summed overload percentage points + 0.05(positive loss increase percent).

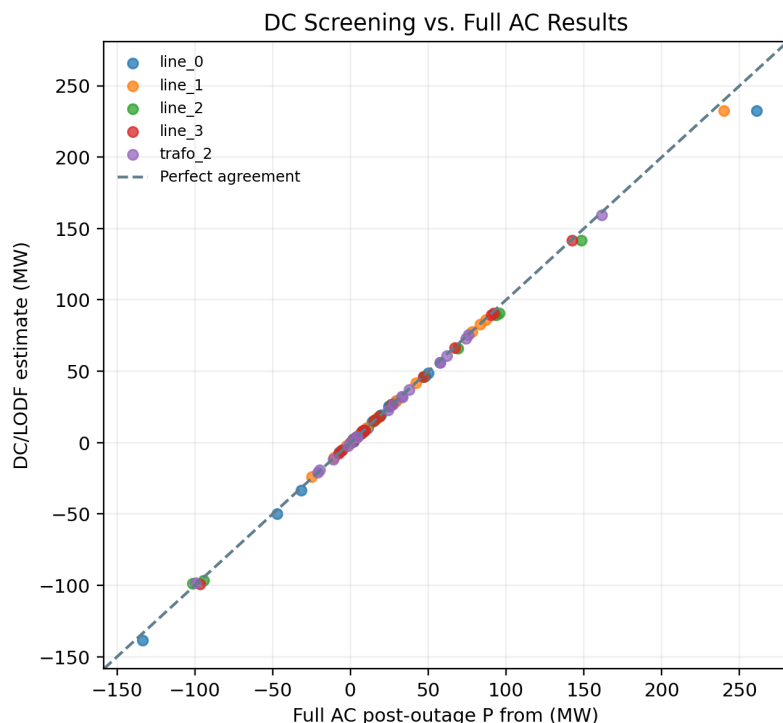
Highest-ranked outages

Rank	Outage	Status	Max loading	V violations	Loss MW	Score
1	T04 7-8	Island	-	0	-	1750.0
2	L01 1-2	Solved	277.6%	2	41.97	349.7
3	L03 2-3	Solved	263.8%	2	24.74	291.7
4	T03 5-6	Solved	148.7%	6	16.68	178.0
5	L02 1-5	Solved	141.8%	2	21.00	117.4



DC/LODF Screening After Full AC Analysis

The lossless DC model forms PTDFs from the reduced nodal susceptance matrix. For outage k , $\text{LODF}[m,k] = h[m,k] / (1 - h[k,k])$ and $F^m = F_m + \text{LODF}[m,k]F_k$. Transformer 7-8 is a bridge, producing a singular denominator; the screen marks it unscreenable rather than returning a false estimate.



Selected-case comparison

Five highest-ranked connected outages

Overall MAE: 1.174 MW

PTDF reference difference:
2.0e-15

LODF reference difference:
6.5e-15

The largest discrepancy occurs after the heavy line 1-2 outage, where AC loss and voltage effects matter more.

Where the approximation succeeds and fails

DC/LODF efficiently estimates active-flow redistribution for connected, moderate-angle cases. It cannot assess reactive power, voltage limits, real losses, generator Q limits, nonlinear convergence, or valid behavior after islanding. Full AC results remain authoritative.

Verification

Seven automated tests verify base convergence and MW balance, exactly 20 unique contingencies, no missing IDs, voltage/thermal boundary logic, deterministic ranking, bridge handling and LODF diagonals, and agreement between the independent PTDF/LODF implementation and PYPOWER.

Limitations and next steps

Replace synthetic ratings with normal/emergency utility ratings; reconcile voltage criteria with PV setpoints; enforce generator reactive limits and PV-to-PQ switching; add seasonal operating cases, remedial actions, and production-tool benchmarking. This teaching model supports method demonstration, not grid operation.

Primary references

MATPOWER case14: matpower.org/docs/ref/matpower5.0/case14.html

MATPOWER PTDF/LODF: matpower.org/documentation/ref-manual/legacy/functions/makePTDF.html and [makeLODF.html](http://matpower.org/documentation/ref-manual/legacy/functions/makeLODF.html)

Pandapower: pandapower.readthedocs.io